



The plant immune system in heterogeneous environments

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The plant immune system inhibits pathogen growth and contributes to shaping a healthy microbial community in the plant body. Plants must appropriately respond to both microbial signals and abiotic factors that are diverse in time and space, and thus, proper integration of these inputs at local and systemic levels is of crucial importance for optimal plant responses and fitness in nature. Here, we review our current knowledge of three properties of the plant immune system, resilience, tunability, and balance, which enable plants to deal with complex cocktails of environmental factors. We also discuss future challenges on the path towards a comprehensive understanding of the interactions between plant immunity and pathogenic, non-pathogenic, and beneficial microbes.

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Introduction

The ultimate role of plant immunity, from the human point of view, is to stop off all potential pathogens that endanger plant survival. One straightforward way to accomplish this might be to devise parallel and constitutively active immune branches that are collectively effective against all pathogens. However, it appears that evolution did not favor this way for several reasons. First, the resources available to plants are limited. Plants need to allocate a finite amount of resources to their growth, to responses to abiotic stresses, as well as to the fight against pathogens; thus, constitutively active immunity poses a burden on plant fitness. Second, there is no immune response sufficiently versatile to be effective against all pathogens and some immune responses even lead to opposing outcomes in pathogen growth. For instance,

hypersensitive cell death is an immune response effective against some biotrophic pathogens, which rely on living host cells, but it promotes the growth of necrotrophic pathogens, which feed on dead cells [1]. In addition, constitutively active immune responses affect the structure of plant-associated microbial communities (microbiota) [2], which may be deleterious to plant health. In mammals, it is known that excessive activation of host immunity can lead to abnormal microbiota structure, which negatively affects host health [3].

An inducible immune system might help plants to reconcile these problems. Plants have evolved immune sensors such as pattern recognition receptors (PRRs) [4] and nucleotide-binding domain and leucine-rich repeat-containing proteins (NLRs) [5] that are activated upon contact with pathogens. These receptors recognize a variety of ligands including microbial/pathogen-associated molecular patterns (MAMPs/PAMPs) [4,6], pathogen molecules called effectors that are translocated from pathogens into plants to interfere with plant immunity [7], and damage-associated molecular patterns (DAMPs) that signal damaged/alter self [4,6]. Pattern-triggered immunity (PTI) and effector-triggered immunity (ETI) are the major types of plant immunity triggered by MAMPs and effectors, respectively [8]. An energy-efficient and flexible immune system would be advantageous for plant survival and, therefore, be selected. Indeed, plants are able to (i) deploy immune responses that are resilient against perturbations by biotic and abiotic factors, (ii) fine-tune immune responses in context-dependent manners, and (iii) prioritize or balance immune responses in relation to other responses. Such traits can be explained by interwoven plant immune networks comprised of immune receptors, plant hormones, transcription factors, and other components [9].

Different tissues or cells within a plant are exposed to different sets of environmental inputs due to the inherent heterogeneity of environmental factors in time and space. Thus, it is likely that plant immune networks have evolved to integrate such heterogeneous inputs and respond in context-dependent manners. Understanding how plant systems respond to heterogeneous environments is crucial for engineering crops with increased yields. In this review, we summarize our knowledge of the plant immune system focusing on its resilient, tunable, and balanced properties and discuss roles of the immune system in interactions with pathogens and the microbiota.